

ADITYA DNYANESHWAR DAHALE

Rust & Systems Software Engineer | Contributor to Tokio (Merged), Astral uv, Google, FoxIO

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EDUCATION

Savitribai Phule Pune University (SPPU) — Pune, Maharashtra, India

Expected Graduation: 2029

Bachelor of Engineering (B.E.) in Computer Engineering | 1st Sem SGPA: 8.27 • 2nd Sem SGPA: 8.36

OPEN-SOURCE SYSTEMS CONTRIBUTIONS (TIER-1 REPOSITORIES)

- **tokio-rs / tokio (#8405)** (Asynchronous Runtime for Rust (27,000+ ★)) ✓ Merged PR
Resolved integer overflow & permit-leak in SemaphorePermit::merge (#8395) by transitioning internal accounting from u32 to usize with checked arithmetic, reviewed and merged into master by Tokio lead maintainer.
- **astral-sh / uv (#21457)** (Fast Python Package & Project Manager in Rust (45,000+ ★)) ✓ PR Green (55 CI Passed)
Refactored cross-process lock timeouts into EnvironmentOptions (#14720), ensuring thread-safe atomic lock acquisition across 11 OS/architecture matrices (Linux, Windows, macOS, Android, FreeBSD) with zero Clippy/Hawk linter warnings.
- **google / comprehensive-rust (#3275)** (Google's Official Rust Training Courseware (28,000+ ★)) ✓ PR Green (CLA Signed)
Accurately modeled Undefined Behavior in unsafe Rust courseware using i32::unchecked_add (#3122), aligning instruction with the Rust memory model per Google Staff Engineer design review.
- **FoxIO-LLC / ja4 (#312 & #314)** (Industry Standard Network/TLS Fingerprinting in Rust) ✓ 2x Merged PRs
Eliminated heap allocations during HTTP/1 header filtering (#312); optimized hash12 by direct hex-encoding 6-byte digest slices in ja4/ja4x, removing 2 allocations and avoiding encoding 26 unused bytes (#314).
- **smoltcp-rs / smoltcp (#1194)** (Embedded Zero-Copy TCP/IP Network Stack in Rust) ✓ Merged PR
Engineered a single-pass zero-allocation TcpOptionsSummary parser, eliminating dynamic memory allocations on the packet wire-path to accelerate packet ingestion.

FEATURED SYSTEMS & AI PROJECTS

Spryzen+ — High-Throughput Network Inspection Engine & AI Gateway

Rust, eBPF/XDP, JA4+ Telemetry, Docker

- Engineered zero-allocation network telemetry & JA4 TLS inspection engine achieving **4.83 Million RPS with 0.207µs P99 latency** on bare-metal (reproducible via public Docker test harness).
- Implemented single-pass Client Hello parser to extract deterministic JA4 fingerprints without full TLS decryption.
- Architected sovereign AI gateway dashboard (Next.js, WebGL) with real-time attack telemetry and JWT/2FA session auth.

EcoSwap 2.0 — AI-Powered Sustainable E-Commerce Platform

Python, Django, Gemini API, RAG, Web Speech

- Engineered Retrieval-Augmented Generation (RAG) semantic search across 600+ catalog items with Gemini LLM fallbacks to prevent hallucination.
- Integrated Web Speech API for voice queries and real-time environmental sustainability impact analytics.

TECHNICAL SKILLS

- **Systems & Rust:** Rust (Tokio, Async, Concurrency, Zero-Copy Parsing), eBPF/XDP, Network Protocols (TCP/IP, TLS, JA4), Memory Safety
- **Backend & AI:** Python (FastAPI, Django), LiteLLM, Pydantic, RAG, LangChain, REST APIs, Webhooks, Docker, Linux, Git/GitHub
- **Databases & Web:** PostgreSQL, Redis, SQLite, Next.js, React, Tailwind CSS, TypeScript, Three.js / WebGL

VERIFIED CERTIFICATIONS

- **AWS Cloud Practitioner Essentials** — Amazon Web Services | **AWS AI Fundamentals & Cloud** — AWS / Coursera
- **Developing AI Applications with Python & Flask** — IBM | **Python for Data Science & AI** — IBM
- **Microsoft Python Programming Fundamentals** — Microsoft | **Generative AI Evolution** — LinkedIn Learning